

MIXED HODGE STRUCTURES AND EQUIVARIANCE ON SIMPLICIAL COHOMOLOGY

Let $X_\bullet$ be a smooth projective simplicial scheme, meaning that each X_p is a finite disjoint union of smooth projective complex varieties. Let $f_\bullet : X_\bullet \rightarrow X_\bullet$ be a simplicial endomorphism, with degree- p map $f_p : X_p \rightarrow X_p$. All cohomology refers to the associated complex analytic spaces. We first construct the two filtrations on simplicial cohomology.

Definition 1 (Construction of the weight and Hodge filtrations). Write $d_i^{(p)} : X_p \rightarrow X_{p-1}$ for the face maps. Set

$$K^{p,q} := C_{\text{sing}}^q(X_p^{\text{an}}, \mathbb{Q}) \quad (p, q \geq 0), \quad K^{p,q} := 0 \quad \text{otherwise,}$$

and define

$$h|_{K^{p,q}} = \sum_{i=0}^{p+1} (-1)^i (d_i^{(p+1)})^*, \quad v|_{K^{p,q}} = (-1)^p d_{\text{sing}},$$

$$C^m := \text{Tot}^m K = \bigoplus_{p+q=m} K^{p,q}, \quad D = h + v.$$

The simplicial identities give $h^2 = v^2 = hv + vh = 0$. Thus $D^2 = 0$, and $H^n(X_\bullet, \mathbb{Q}) = H^n(C, D)$. The column filtration is

$$L^p C^m := \bigoplus_{i \geq p} K^{i, m-i}, \quad L^p H^n(C) := \text{im}(H^n(L^p C) \longrightarrow H^n(C)).$$

Define the increasing weight filtration in total degree n by

$$W_w H^n(X_\bullet, \mathbb{Q}) := L^{n-w} H^n(C) = \text{im}(H^n(L^{n-w} C) \longrightarrow H^n(C)).$$

Thus a class lies in $W_w H^n$ precisely when it has a closed total representative supported in columns $i \geq n - w$.

For the Hodge filtration, let $\mathcal{A}^{a,b}(X_p)$ denote the space of smooth complex-valued differential forms of type (a, b) on X_p^{an} . Form the total complex

$$B^m := \bigoplus_{p+a+b=m} \mathcal{A}^{a,b}(X_p), \quad D_B = h + (-1)^p d_{\text{dR}},$$

where all indices are nonnegative and h is again the alternating sum of face pullbacks. Define its decreasing filtration by

$$F^\ell B^m := \bigoplus_{\substack{p+a+b=m \\ a \geq \ell}} \mathcal{A}^{a,b}(X_p).$$

Holomorphic pullbacks preserve type, and $d_{\text{dR}} = \partial + \bar{\partial}$ does not decrease the first type index a . Hence $D_B(F^\ell B) \subseteq F^\ell B$. The natural de Rham comparison gives

$$c_n : H^n(B, D_B) \xrightarrow{\sim} H^n(X_\bullet, \mathbb{Q}) \otimes_{\mathbb{Q}} \mathbb{C} =: H^n(X_\bullet, \mathbb{C}).$$

Under this comparison, define

$$F^\ell H^n(X_\bullet, \mathbb{C}) := c_n \left(\text{im} \left(H^n(F^\ell B) \longrightarrow H^n(B) \right) \right).$$

Thus $F^\ell H^n$ consists of classes admitting a closed total form representative whose components all have $a \geq \ell$.

Deligne's construction, applied with empty boundary in every smooth projective level, proves that the filtrations just defined make

$$(H^n(X_\bullet, \mathbb{Q}), W, F)$$

a mixed Hodge structure [D74, (8.1.19)–(8.1.20)]. Levelwise pullback by f_p preserves columns on C and types on B , and the de Rham comparison is natural. Consequently $\Phi = f_\bullet^*$ preserves W and F and is an endomorphism of this mixed Hodge structure. We now describe its induced actions on the associated graded spaces.

Definition 2 (Filtrations and associated graded spaces). An increasing finite filtration on a rational vector space V is a family of subspaces

$$\cdots \subseteq W_{w-1}V \subseteq W_wV \subseteq W_{w+1}V \subseteq \cdots,$$

with $W_wV = 0$ for $w \ll 0$ and $W_wV = V$ for $w \gg 0$. Its associated graded spaces are the quotients

$$\mathrm{Gr}_w^W V := W_wV / W_{w-1}V.$$

A decreasing finite filtration L has $L^{p+1}V \subseteq L^pV$, with $L^pV = V$ for $p \ll 0$ and $L^pV = 0$ for $p \gg 0$, and

$$\mathrm{Gr}_L^p V := L^pV / L^{p+1}V.$$

For $V = H^n(X_\bullet, \mathbb{Q})$, the MHS also has a decreasing Hodge filtration F on $V_{\mathbb{C}} = V \otimes_{\mathbb{Q}} \mathbb{C}$. On $(\mathrm{Gr}_w^W V)_{\mathbb{C}}$ the induced filtration is

$$F^\ell(\mathrm{Gr}_w^W V)_{\mathbb{C}} = \frac{(F^\ell V_{\mathbb{C}} \cap W_w V_{\mathbb{C}}) + W_{w-1} V_{\mathbb{C}}}{W_{w-1} V_{\mathbb{C}}}.$$

This is a pure Hodge structure of weight w .

An endomorphism Φ of the MHS V satisfies

$$\Phi(W_wV) \subseteq W_wV, \quad \Phi_{\mathbb{C}}(F^\ell V_{\mathbb{C}}) \subseteq F^\ell V_{\mathbb{C}}.$$

It therefore induces an endomorphism of each weight-graded Hodge structure by the explicit quotient rule

$$\mathrm{Gr}_w^W(\Phi)(\alpha + W_{w-1}V) = \Phi(\alpha) + W_{w-1}V \quad (\alpha \in W_wV).$$

Indeed, changing α by an element of $W_{w-1}V$ changes $\Phi(\alpha)$ by an element of that same subspace. By construction, in degree n we have

$$L^pV = W_{n-p}V.$$

Then $\mathrm{Gr}_L^p V = \mathrm{Gr}_{n-p}^W V$, and the induced map is

$$\Phi_L^p := \mathrm{Gr}_L^p(\Phi), \quad \Phi_L^p(\alpha + L^{p+1}V) = \Phi(\alpha) + L^{p+1}V.$$

Thus the action on $\mathrm{Gr}_L^p V$ is already determined by the MHS endomorphism Φ .

Define $T : C \rightarrow C$ by $T|_{K^{p,q}} = f_p^*$. The identities $d_i^{(p)} f_p = f_{p-1} d_i^{(p)}$ imply $Th = hT$, and naturality of the singular differential gives $Tv = vT$. Consequently $TD = DT$. Since T preserves every bidegree, in particular $T(L^pC) \subseteq L^pC$. It is therefore a filtered cochain endomorphism, and

$$\Phi = f_\bullet^* = H^n(T), \quad \Phi([a]_D) = [Ta]_D.$$

Here $[a]_D$ denotes the cohomology class of a closed total cochain. The MHS condition alone gives the action Φ_L^p on the graded space. The cochain realization $\Phi = H^n(T)$ relates it to the spectral-sequence actions described next.

The column filtration gives the first-quadrant spectral sequence

$$\begin{aligned} E_1^{p,q} &= H^q(X_p, \mathbb{Q}), \quad d_1^{p,q} = \delta^{p,q}, \\ E_2^{p,q} &= \frac{\ker \delta^{p,q}}{\operatorname{im} \delta^{p-1,q}}, \quad E_1^{p,q} \implies H^{p+q}(X_\bullet, \mathbb{Q}), \end{aligned}$$

where

$$\delta^{p,q} := \sum_{i=0}^{p+1} (-1)^i (d_i^{(p+1)})^* : H^q(X_p, \mathbb{Q}) \longrightarrow H^q(X_{p+1}, \mathbb{Q}),$$

and $\operatorname{im} \delta^{-1,q} = 0$. Filtered-complex functoriality gives endomorphisms $T_r^{p,q} : E_r^{p,q} \rightarrow E_r^{p,q}$ satisfying

$$d_r T_r = T_r d_r, \quad T_{r+1} = H(T_r)$$

[W94, Construction Theorem 5.4.1, p. 132]. Their construction does not require convergence. In particular,

$$T_1^{p,q} = f_p^*, \quad T_2^{p,q}(\alpha + \operatorname{im} \delta^{p-1,q}) = f_p^* \alpha + \operatorname{im} \delta^{p-1,q}.$$

The latter formula is well-defined because $f_{p+1}^* \delta^{p,q} = \delta^{p,q} f_p^*$ preserves kernels and, using the preceding degree, images.

Theorem 1 (Equivariant comparison maps). *For $p, q \geq 0$ and $n = p + q$, convergence and degeneration give canonical isomorphisms of pure Hodge structures of weight q ,*

$$\rho_{p,q} : \operatorname{Gr}_L^p H^n(X_\bullet, \mathbb{Q}) \xrightarrow{\sim} E_\infty^{p,q}, \quad \sigma_{p,q} : E_2^{p,q} \xrightarrow{\sim} E_\infty^{p,q}.$$

They satisfy

$$\rho_{p,q} \Phi_L^p = T_\infty^{p,q} \rho_{p,q}, \quad \sigma_{p,q} T_2^{p,q} = T_\infty^{p,q} \sigma_{p,q}.$$

Consequently

$$\alpha_{p,q} := \sigma_{p,q}^{-1} \rho_{p,q} : \operatorname{Gr}_L^p H^n(X_\bullet, \mathbb{Q}) \xrightarrow{\sim} \frac{\ker \delta^{p,q}}{\operatorname{im} \delta^{p-1,q}}$$

is equivariant, with the quotient action induced by f_p^* :

$$\begin{array}{ccc} \operatorname{Gr}_L^p H^n(X_\bullet, \mathbb{Q}) & \xrightarrow[\sim]{\alpha_{p,q}} & \frac{\ker \delta^{p,q}}{\operatorname{im} \delta^{p-1,q}} \\ \Phi_L^p \downarrow & & \downarrow T_2^{p,q} \\ \operatorname{Gr}_L^p H^n(X_\bullet, \mathbb{Q}) & \xrightarrow[\sim]{\alpha_{p,q}} & \frac{\ker \delta^{p,q}}{\operatorname{im} \delta^{p-1,q}}. \end{array}$$

Proof. The column filtration is finite in every degree. The convergence isomorphism therefore gives $\rho_{p,q} : \operatorname{Gr}_L^p H^n(C) \xrightarrow{\sim} E_\infty^{p,q}$ [W94, Classical Convergence Theorem 5.5.1(1), p. 135]. Each $E_1^{p,q} = H^q(X_p, \mathbb{Q})$ is pure of weight q . The maps $\delta^{p,q}$ are morphisms of Hodge structures, so $E_2^{p,q}$ is also pure of weight q . Every higher differential is a morphism of Hodge structures [D74, Proposition (8.3.5)]. For $r \geq 2$, induction shows that its source and target

$$d_r^{p,q} : E_r^{p,q} \longrightarrow E_r^{p+r, q-r+1}$$

have distinct pure weights q and $q - r + 1$, respectively. Thus $d_r^{p,q} = 0$ for every $r \geq 2$, and the canonical identifications $E_r \cong E_{r+1}$ give $\sigma_{p,q} : E_2^{p,q} \xrightarrow{\sim} E_\infty^{p,q}$. Compatibility with the Hodge filtration makes $\rho_{p,q}$ and $\sigma_{p,q}$ isomorphisms of pure Hodge structures [D74, (8.1.20)(ii)].

For equivariance, let $a \in L^p C^n$ be closed and put $\bar{a} = [a]_D + L^{p+1} H^n(C)$. Since $\Phi = H^n(T)$, both compositions send $\bar{a}$ to the stable class of Ta :

$$\rho_{p,q} \Phi_L^p(\bar{a}) = [Ta]_\infty = T_\infty^{p,q} \rho_{p,q}(\bar{a}).$$

Likewise, for $z \in E_2^{p,q}$, degeneration identifies $\sigma_{p,q}(z)$ with its stable class z_∞ . Since $T_{r+1}([z]) = [T_r z]$ on every page,

$$\sigma_{p,q}(T_2^{p,q} z) = (T_2^{p,q} z)_\infty = T_\infty^{p,q} \sigma_{p,q}(z).$$

Consequently $\alpha_{p,q} = \sigma_{p,q}^{-1} \rho_{p,q}$ is equivariant:

$$\alpha_{p,q} \Phi_L^p = \sigma_{p,q}^{-1} T_\infty^{p,q} \rho_{p,q} = T_2^{p,q} \alpha_{p,q}.$$

□

For a concrete description, write a closed representative as

$$a = \sum_{i=p}^n a_i \in L^p C^n, \quad a_i \in K^{i, n-i}.$$

The equation $Da = 0$ gives $va_p = 0$ and $ha_p + va_{p+1} = 0$ (with $a_{p+1} = 0$ when $q = 0$), so the singular cohomology class $[a_p]_{\text{sing}}$ belongs to $\ker \delta^{p,q}$. The comparison is

$$\alpha_{p,q}([a]_D + L^{p+1} H^n) = [a_p]_{\text{sing}} + \text{im } \delta^{p-1,q}.$$

Indeed, ρ assigns the stable class of a , and σ sends the displayed E_2 -class to that same stable class. Convergence supplies ρ ; degeneration at E_2 supplies σ and its inverse. Convergence alone does not identify E_∞ with E_2 .

REFERENCES

- [D74] P. Deligne, *Théorie de Hodge, III*, Publ. Math. Inst. Hautes Études Sci. **44** (1974), 5–77. [\(8.1.19\)–\(8.1.20\)](#), pp. 34–35; [Proposition 8.3.5](#), p. 42.
- [W94] C. A. Weibel, *An Introduction to Homological Algebra*, Cambridge Studies in Advanced Mathematics, vol. 38, Cambridge University Press, 1994. [Construction Theorem 5.4.1](#), p. 132; [Classical Convergence Theorem 5.5.1](#), p. 135. For cohomological indexing, see Definition 5.2.3, p. 123.